Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Everything I've learned in this class so far will help me complete this project. I expect to use loops and lists, nested dataclasses, functions, and others. I want to improve on using drafter and nested dataclasses.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I feel like creating the dataclasses will be the easiest part because I understand them the most and I think it will be easy to organize my thoughts that way. The hardest part will probably be creating the drafter part of the program because I don't really understand drafter and the actual functions I need to sort the books and return one that matches the user's input.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think the most time consuming tasks will be creating the functions that filter books based on the user's preference and making sure the Drafter website works correctly. I should implement the Book dataclass and create a list of books first. Then I will work on the functions that filter the books and randomly select a recommendation.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

My phone is my biggest distraction, also my procrastination and other assignments will cause some distractions. I will minimize them by giving my phone to my roommate while I do my work and find a quiet place to work. I will use my class notes, my computer, and help from my professor or classmates if I get stuck. I have no questions at the moment.

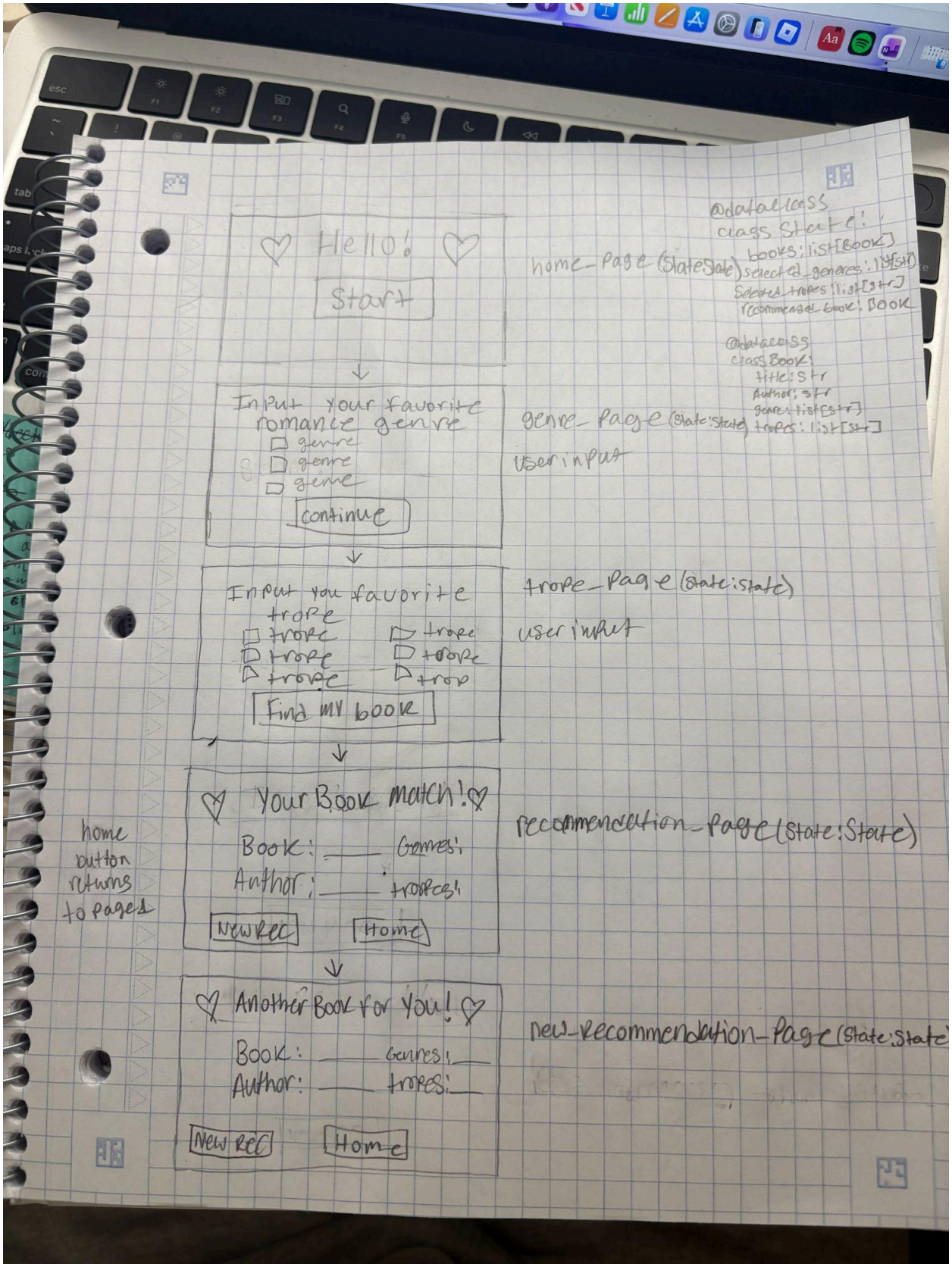
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Hello!
Start
Input your favorite romance genre
genre
genre
genre
continue
Input you favorite trope
trope
trope
trope
trope
trope
trop
Find my book
Your Book match!
Book: ____ Genres:
Author: ____ tropes:
NewRec
Home
Another Book for You!
Book: ____ Genres: ____
Author: ____ tropes: ____
New Rec
Home
home button returns to pages
@dataclass
class State:
books: list[Book]
selected_genres: list[str]
selected_tropes: list[str]
recommended_book: Book
@dataclass
class Book:
title: str
Author: str
genres: list[str]
tropes: list[str]
home_page(State:State)
genre_page(State:State)
user input
trope_page(State:State)
user input
recommendation_page(State:State)
new_recommendation_page(State:State)

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	My website will help indecisive readers discover romance books based on their favorite genres and tropes. Users will input their preferences, and the website will randomly choose a matching book and display its title, author, genres, and tropes.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	My State dataclass will contain 4 attributes. The first is a list of Book instances called books, which stores all the romance books available for recommendations. The second is selected_genres, a list of strings that stores the genres selected by the user. The third is selected_tropes, another list of strings that stores the users favorite tropes. The fourth is recommended_book, which stores the Book instance that was randomly selected. The attributes will be updated as users make selections and receive new recommendations.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	My second dataclass will be called Book. it will contain 4 attributes: title(str), author(str), genres(list[str]), and tropes(list[str]). Each Book instance will represent one romance book and store the information needed to compare it with the user's preferences.
Routes (minimum 7) The names of your route functions	home_page() genre_page() trope_page() preferences_page() recommendation_page()

	new_recommendation_page() reset_preferences()
Pages (minimum 5) A brief description of the pages in your site	The first page will be a Home page that welcomes users and has a Start button. The second page will allow users to select their favorite romance genres. The third page will allow users to select their favorite romance tropes. The fourth page will display the first recommended book, including its title, author, genres, and tropes. It will also have buttons to return to the Home page or get a new recommendation. The fifth page will display a new book recommendation and allow users to get another recommendation or return to the Home page.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	My website will use checkboxes to allow users to select their favorite romance genres, such as contemporary, fantasy, and historical. It will also use checkboxes to let users select their favorite tropes, such as friends to lovers, enemies to lover, and fake dating. These inputs will save the user's selections in the State dataclass so the website can recommend books that match their preferences.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	My website will use if statement to check whether the user selected any genres, whether the users selected any tropes, and whether a book matches the user's preferences. If the user does not select any preferences, the website can ask them to choose some before continuing. If a book matches the selected genres and tropes, it can be added to the list of possible recommendations. If no books match, the website will display a message asking the user to change their preferences.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in	My website will use a for loop to go through the list of Book instances stored in the State dataclass,. The loop will compare each book's genres and tropes

your State attribute. Description of this loop and why it is meaningful to the site	with the user's selected preference. Books that match the preferences will be added to a list of possible recommendations. This will allow the website to find books that fit what the user is looking for.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

i have 6 out of 7 routes for my page i need another one here is what im working on

My website will help indecisive readers discover romance books based on their favorite genres and tropes. Users will input their preferences, and the website will randomly choose a matching book and display its title, author, genres, and tropes.

My State dataclass will contain 4 attributes. The first is a list of Book instances called books, which stores all the romance books available for recommendations. The second is selected_genres, a list of strings that stores the genres selected by the user. The third is selected_tropes, another list of strings that stores the users favorite tropes. The fourth is recommended_book, which stores the Book instance that was randomly selected. The attributes will be updated as users make selections and receive new recommendations. can you help me with another route

Routes (minimum 7)	home_page()
	genre_page()
The names of your route functions	trope_page()
	recommendation_page()
	new_recommendation_page()
	reset_preferences()

Absolutely! I think a good seventh route for your website would be:

`preferences_page()`

This route will display the user's currently selected genres and tropes so they can review their preferences before receiving a book recommendation.

You could add it to your table like this:

Routes	Names of your route functions
	`home_page()`
	`genre_page()`
	`trope_page()`
	`preferences_page()`
	`recommendation_page()`
	`new_recommendation_page()`

```
reset_preferences()
```

Why this route works: It gives users a chance to review their selected genres and tropes before the website randomly chooses a matching romance book.

It also makes sense with your `State` dataclass because it uses `selected_genres` and `selected_tropes` to display the user's preferences.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.